

Ashwin Conrad

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Mechanical engineering student focused on vehicle systems, manufacturing, testing, and build-ready design. Experience in Formula EV low-voltage integration, industrial control-panel QA, engineering analysis, and Python automation.

EDUCATION

University of Alberta | Faculty of Engineering, BSc in Mechanical Engineering - Edmonton, AB | **2023 - Present**

West Island College | High School - Calgary | **2020-2023**

- Played varsity Basketball & Rugby; concurrent with Club Basketball and volunteer coaching Youth Basketball.
- Active Leader in the Young Men & Diversity Equity and Inclusion Clubs supporting an inclusive and multicultural community.

CORE CAPABILITIES

Vehicle Systems Integration	Control Panel Assembly & QA	Manufacturing & Test
CAD & Engineering Drawings	Python & Engineering Data	Cross-functional Coordination

EXPERIENCE

Integrity & Reliability Engineering Student 2025 – 2026

AltaGas - Calgary, AB

- Built Python, Microsoft Fabric, and Power BI workflows for corrosion monitoring and inspection planning.
- Analyzed 2+ years of operating data for 12+ brazed aluminum heat exchangers.
- Screened results against OEM limits to identify monitoring gaps and support risk-based recommendations.
- Supported asset-lifecycle work across multi-facility operations.

Control Panel Technician Summer 2023 & 2024

Spartan Controls - Calgary, AB

- Built, wired, loaded, and checked 30+ industrial control panels from drawings and electrical schematics.
- Performed pre-delivery quality checks and documented deficiency resolution before customer delivery.
- Processed 250+ PLC software updates in under half the allocated time while supporting production.
- Supported repeatable assembly, quality, and delivery work in a production environment.

LEADERSHIP

Low Voltage Electronics Lead Winter 2025 - Present

Formula SAE EV - University of Alberta

- Lead the UA-26 Low-Voltage subsystem remotely by focusing on new-member onboarding and transparent documentation.
- Coordinated design goals, integration, and manufacturing with other subsystems to ensure a shared knowledge base.

Engineering Student Engagement Leader 2024-2025

University of Alberta - Edmonton, AB

- As an Engineering Student Engagement Leader, I supported the faculty in hosting open houses, first year events, and community events.

COMMUNITY INVOLVEMENT

Youth Basketball Coach 2023-2026

CYDC - Calgary, AB

- Working with groups of 20-40 kids, ranging in age from 8-17 years old, during weekly practices over 4 month seasons
- Focused player development on teamwork, communication, fundamental basketball skills, agility, and general physical fitness.

RECOGNITION

Undergraduate Student Leadership Award 2024-2025

Leadership in Engineering Scholarship 2023-2024

APEGA Science Olympics Medalist: Silver 2021-2023

PROJECT PORTFOLIO

Industry Projects

Brazed Aluminum Heat-Exchanger Lifecycle Analysis 2025 – 2026

AltaGas - Python, PI Historian Excel add-in

- Analyzed 2+ years of operating data for 12+ brazed aluminum heat exchangers.
- Screened lifecycle evidence against OEM limits and summarized inspection and operating-awareness recommendations.

Inspection Scope Generation Tool 2025 – 2026

AltaGas - Python, openpyxl, python-docx

- Developed a Python tool for equipment-specific inspection-scope generation.
- Generated 500+ standardized scopes from database exports using openpyxl and python-docx.

Course Projects

ENG 160 Mechanical Design Project University Coursework

University of Alberta - Mechanical design process, concept development, technical documentation

- Completed a mechanical design project as part of ENG 160 coursework.
- Developed the design through concept work, evaluation, and technical documentation.

MEC 260 Mechanical Design Project University Coursework

University of Alberta - Mechanical prototyping, packaging, mechanism iteration

- Designed and built a small mechanical vehicle prototype with wheels and linkage.
- Iterated on physical packaging and mechanism performance through hands-on prototyping.

Extracurricular Projects

2026 Low-Voltage Systems and Harness Design, Manufacturing, and Development 2026

Formula EV Racing - Low-voltage electrical systems

- Designed and developed the 2026 low-voltage system and vehicle harness architecture.
- Supported harness manufacturing, integration, serviceability, and cross-team development.

Personal Projects

Leather Tool Carrier 2024

Personal Project - Shop workflow tooling

- Designed a leather tool carrier for common hand tools used during panel assembly.
- Built a durable shop aid that kept frequently used tools organized and accessible.

Composite Skateboard Decks Earlier build series

Personal Project - Lamination, fiberglass layups, embedded aluminum

- Built composite skateboard decks through hands-on lamination and fiberglass layup.
- Experimented with embedded aluminum and finish work across an early build series.

TECHNICAL SKILLS

CAD & DESIGN	SolidWorks, AutoCAD & 2D laser cutting, Engineering drawings, Mechanical prototyping, 3D modelling and printing
VEHICLE & ELECTRICAL	Low-voltage systems, Power distribution, Sensor integration, Electrical schematics, Panel assembly, Pre-delivery QA
DATA & AUTOMATION	Python, Excel data analysis, Power BI, Microsoft Fabric, Power Query, openpyxl, python-docx
ENGINEERING SOFTWARE	Aspen HYSYS, Maximo, JDE
FABRICATION & QA	Engineering drawings, Mechanical prototyping, Panel assembly

Expanded case studies and technical work: ashwin-conrad.github.io